

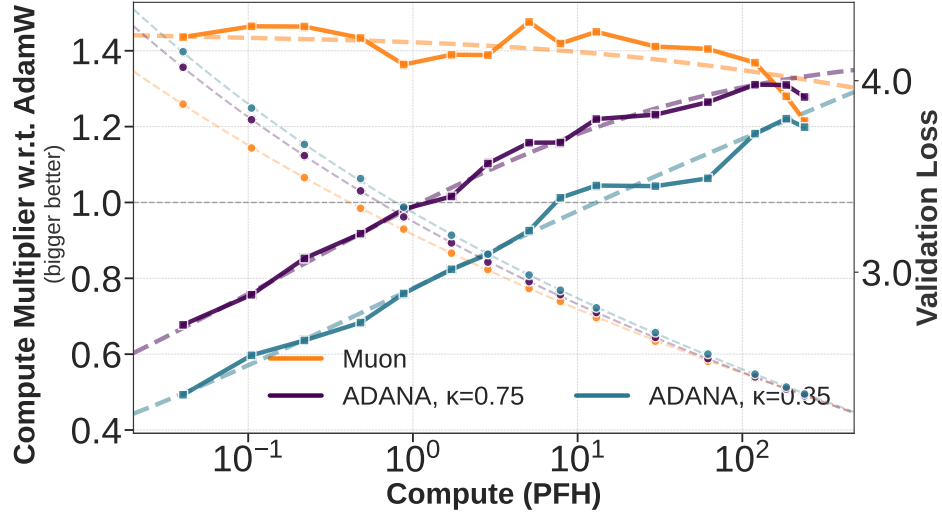


Figure A: **Large Batch $B = 256$ Comparison of Muon, AdamW and ADana with $\alpha(t) = (1 + t)^{1-\kappa}$ and $\kappa \in \{0.75, 0.85\}$.** Under $\alpha(t) = (1 + t)^{1-\kappa}$ with large batch $B = 256$, ADANA with $\kappa = 0.75$ outperforms $\kappa = 0.85$ hence allowing larger damping schedule. The performance increases with scale for ADANA with both $\kappa \in \{0.75, 0.85\}$ eventually outperforming MUON.

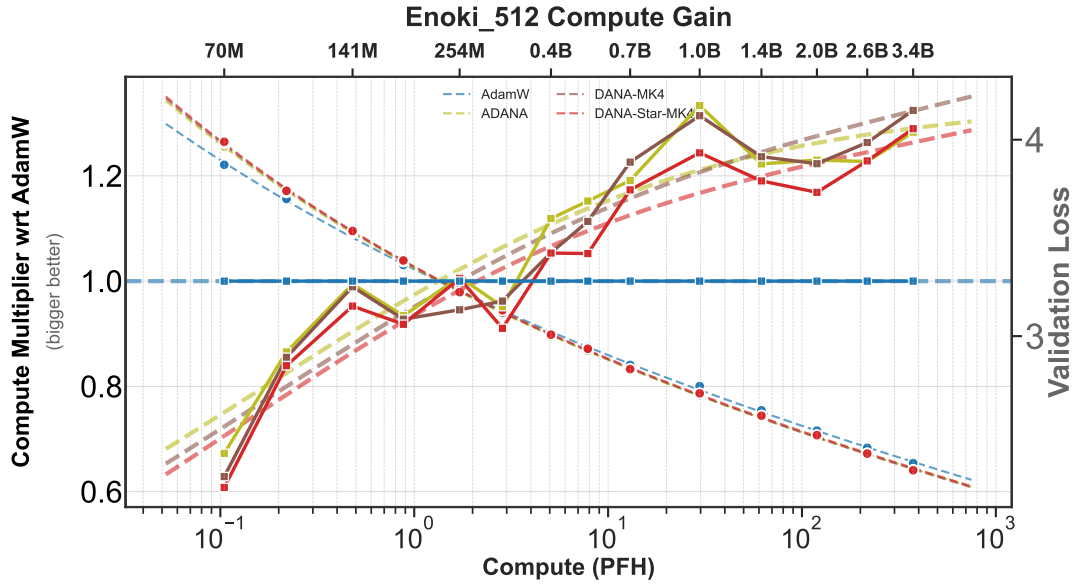


Figure B: **Batch size 512 scaling comparison.** Validation loss comparing ADAMW, ADANA, DANA-MK4, and DANA-STAR-MK4 at batch size 512 with $\tilde{\alpha} = 2.5$.

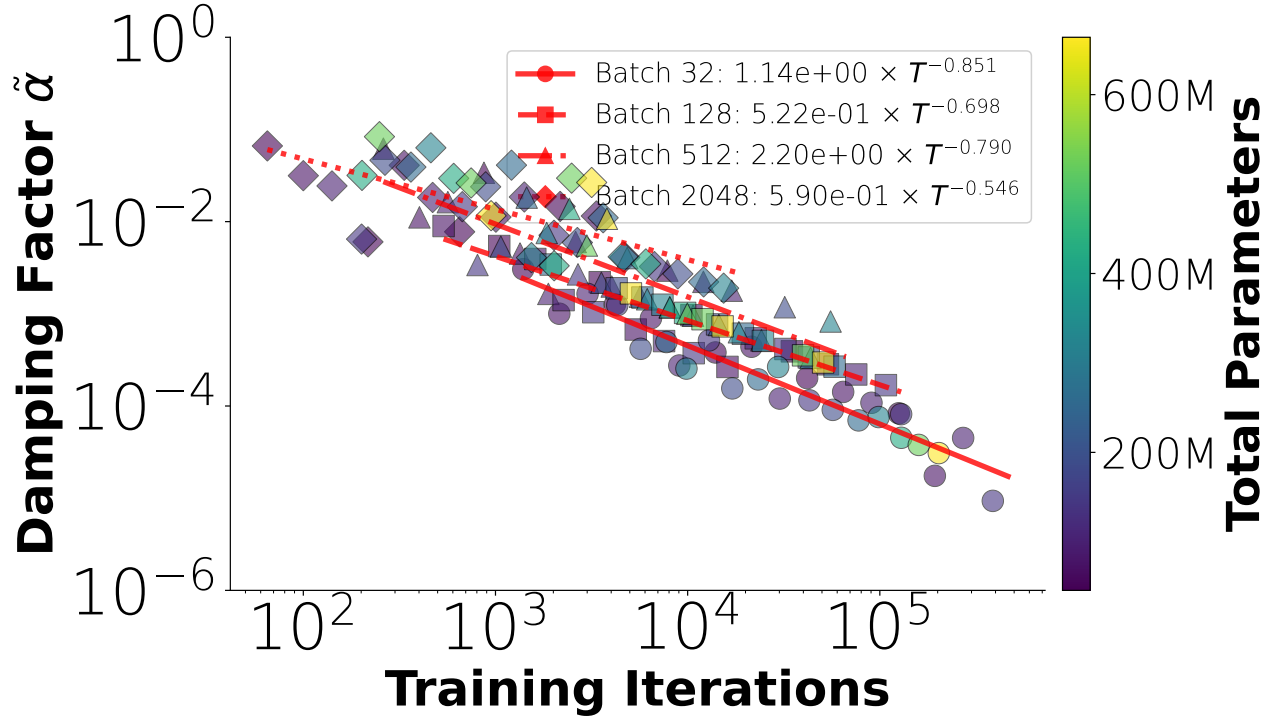


Figure C: **Larger Batch Allow Larger Damping Schedule.** Same setting as in Figure 5: we use $\alpha(t) = \tilde{\alpha} \times (1 + t)$ and sweep $\tilde{\alpha}$ across model size and iterations. Increasing batch size allows for larger constant $\tilde{\alpha}$.

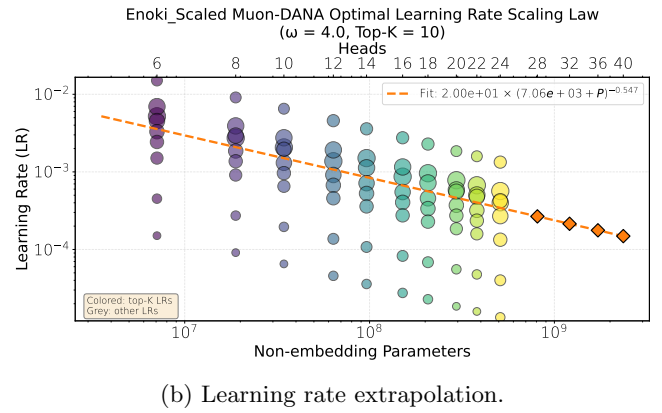
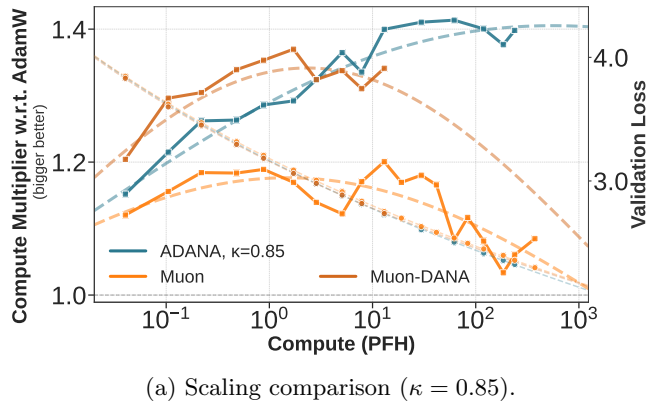
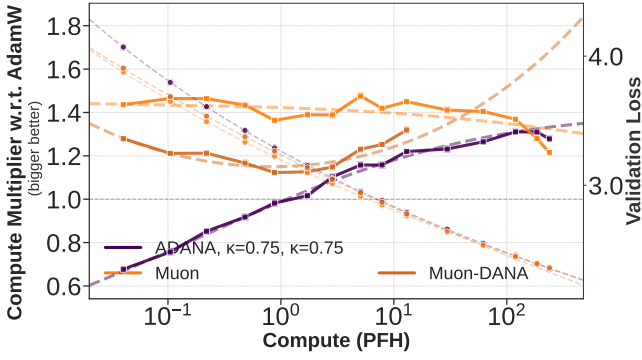
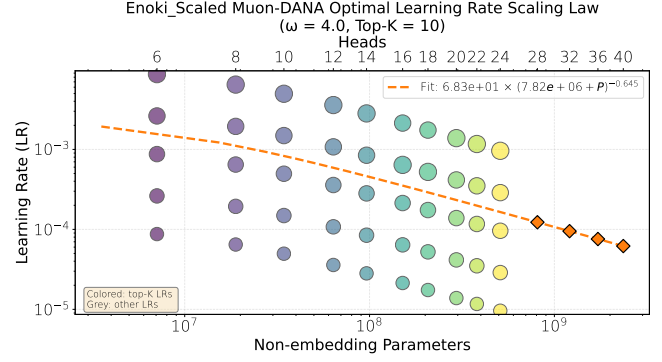


Figure D: **Batch 32**, $\kappa = 0.85$. (a) Scaling comparison of MUON-DANA, ADANA and MUON against AdamW, and (b) learning rate extrapolation.



(a) Scaling comparison ($\kappa = 0.75$, large batch).



(b) Learning rate extrapolation (large batch).

Figure E: **Batch 256**, $\kappa = 0.75$. (a) Scaling comparison of MUON-DANA, ADANA and MUON against AdamW, and (b) learning rate extrapolation.

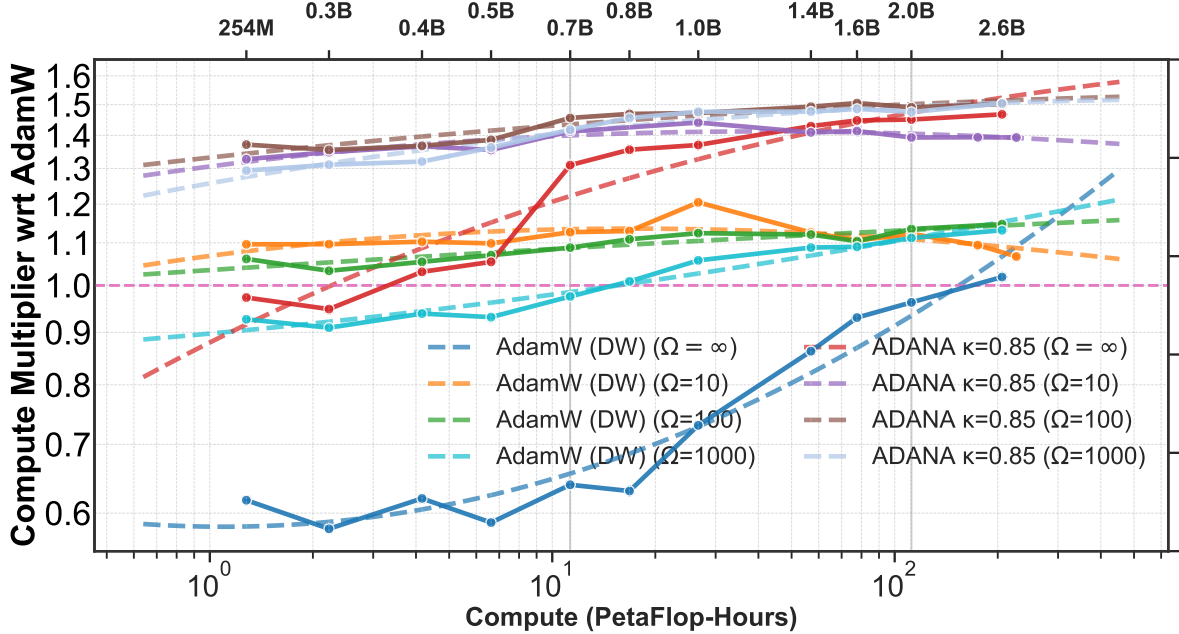


Figure F: **Warm-up in Weight-decay**. We consider a weight-decay schedule of the form $\lambda(t) = \frac{\omega}{T/\Omega+t}$ and vary $\Omega \in \{10, 100\}$, with $\Omega = \infty$ corresponding to $\lambda(t) = \omega/t$. The parameter Ω acts as a warm-up for weight decay, keeping it approximately constant early in training before transitioning to a $1/t$ decay. We find that choosing $\Omega = 10$ or 100 has little effect on the performance of ADANA or ADAMW (DW), while consistently improving performance relative to ADAMW with constant weight decay. In contrast, the performance of ADAMW (DW) degrades substantially when using the pure ω/t schedule, whereas ADANA remains stable under this schedule, at least for larger model sizes but does worse for smaller sizes. It appears that ADANA is less sensitive to the choice of Ω .

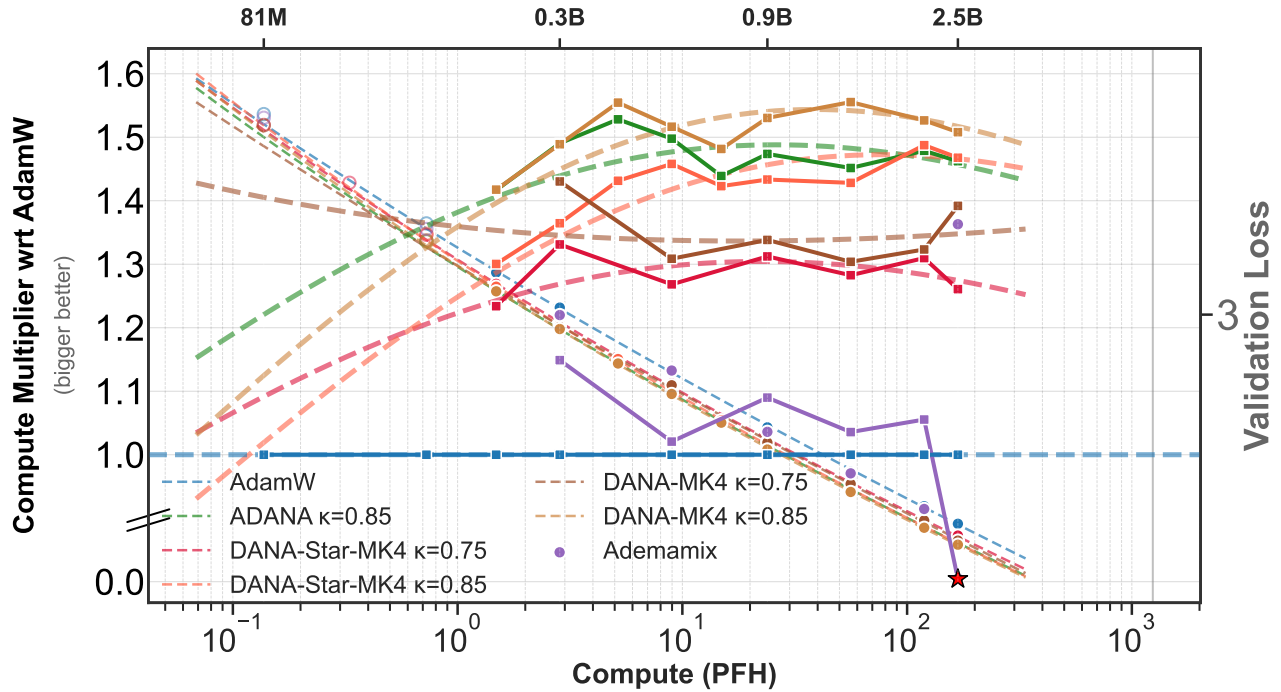
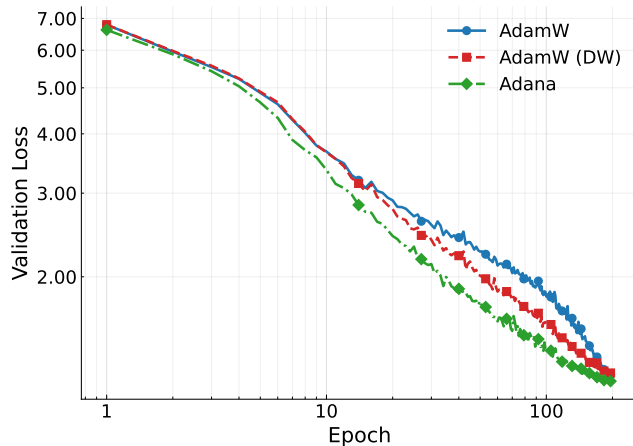
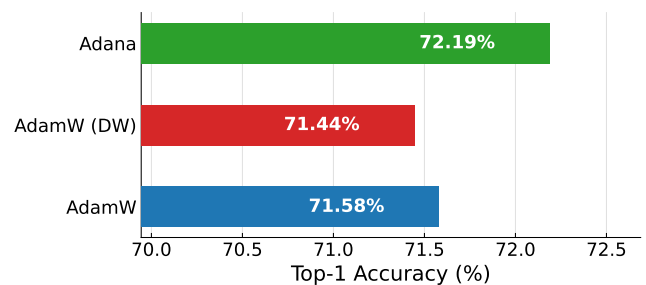


Figure G: **Qwen3 architecture scaling**. *Left*: Compute savings relative to ADAMW. *Right*: Validation loss vs compute with broken power law fits. ADANA ($\kappa = 0.85$), DANA-STAR-MK4 ($\kappa = 0.75, 0.85$), and DANA-MK4 ($\kappa = 0.75, 0.85$) are compared against ADAMW. ADEMAMIX exhibits training instability at 17 heads (marked with star). DANA-MK4 $\kappa = 0.85$ achieves the best loss at most scales.



(a) Validation loss during training.



(b) Final top-1 accuracy.

Figure H: **Vision transformers (ViT) on Imagenet for 200 epochs** Comparison of ADANA, ADAMW and ADAMW with decaying weight-decay (DW). ADANA outperforms ADAMW throughout training and final validation loss. Decaying weight-decay does not seem to improve final validation loss over vanilla ADAMW but smooths out the learning curve. Showing the best learning rate after tuning.

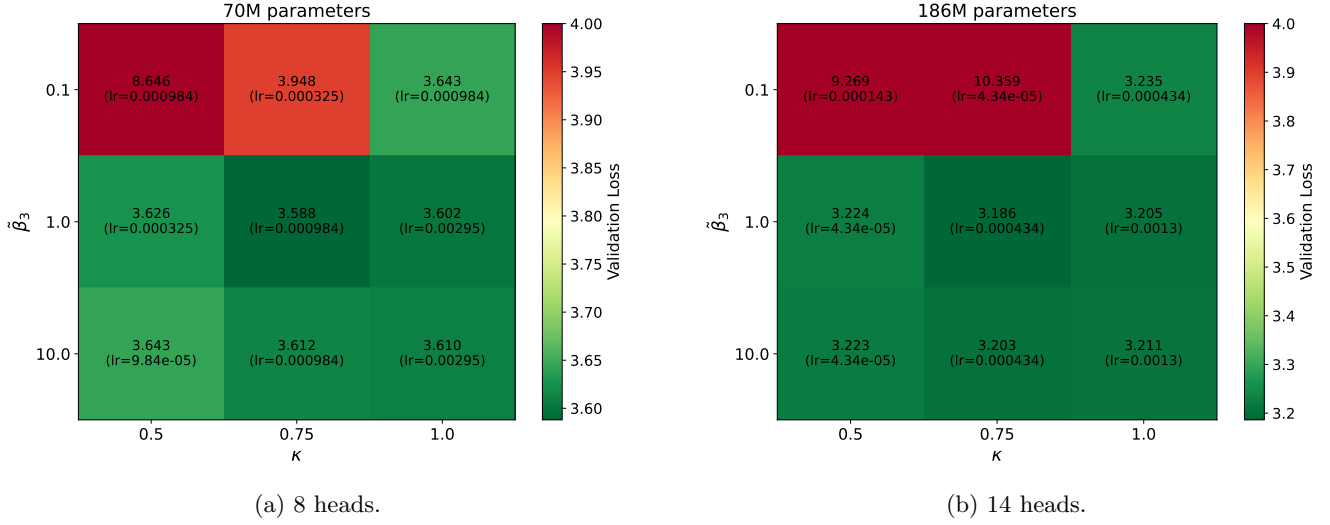


Figure I: **Ademamix hyper-parameters sweep.** Optimal damping exponent κ and long momentum $\beta_3 = 1 - \tilde{\beta}_3 \times \frac{\delta}{T}$ of ADEMAMIX for two different model sizes. Learning rate is tuned over a grid of 5 values with relative ratio 3. In particular, ADANA’s scaling rules $\tilde{\beta}_3 = 1$, $\kappa = 0.75$ is optimal for both sizes.

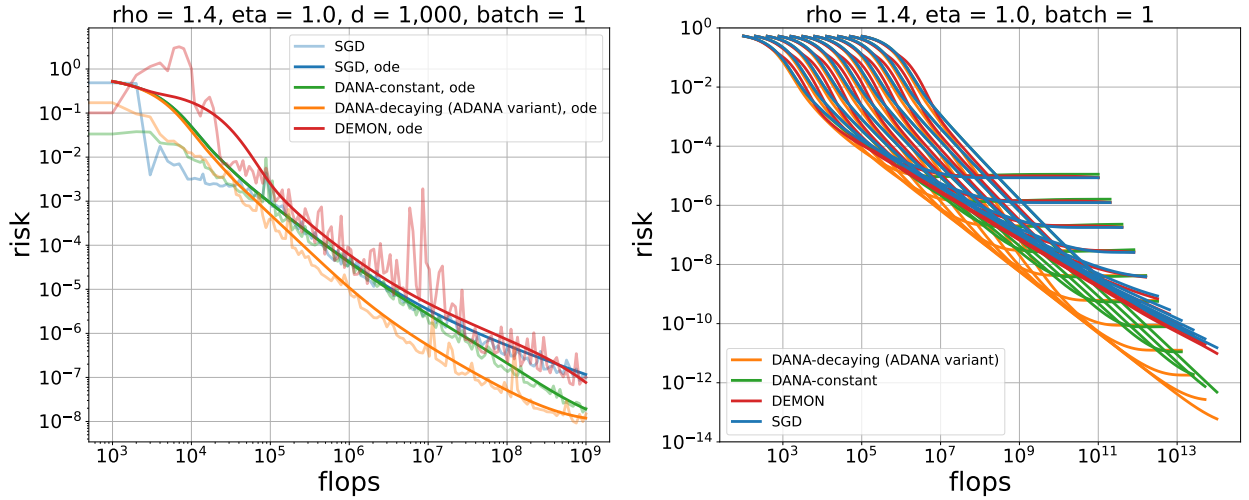


Figure J: **Comparison of SGD, Dana (Alg. 8, Ferbach et al. 2025) with DEMON (Chen et al. 2019)** on the power-law random features model (Sec. B.1) with $\rho = 1.4$ and $\eta = 1.0$. Numerical setup: $D = \text{diag}(j^{-2\alpha} : j = 1, \dots, d)$; SGD (blue curves) learning rate $\gamma = 0.5/\text{tr}(D)$, DANA-constant (green) has $\gamma(t) = 0.5/\text{tr}(D)$, $\alpha(t) = 0.1/d \times (1+t)$, $\beta_1 = 1 - \delta/(1+t)$ where $\delta = \max\{2 - 1/\alpha, (2\alpha + 2\beta - 1)/\alpha\} + 1$; DANA-decaying/ADANA variant (orange) same $\gamma(t)$ and β_1 as DANA-constant, $\alpha(t) = 0.1 \times (1+t)^{1-1/(2\alpha)}$; DEMON (red) $\beta_{\text{init}} = 0.1$ and $\text{lr } \tilde{\eta} = 0.5/\text{tr}(D)$ (Chen et al. 2019); **Left:** algorithms were run for 10^6 steps with $d = 1000$, $v = 10000$; **Right:** algorithms run using the ODEs with equivalences in (Ferbach et al. 2025) since DEMON is a version of Accelerated SGD (?); 10^9 iterations of algorithm, $d = \{100 \times 2^i\}$, $i = 1, \dots, 10$ and $v = 10 \times d$. **DEMON (red) scales very closely with SGD (blue) and not does appear to accelerate** for this (ρ, η) . **We see both Dana-decay (motivation for ADana) and Dana-constant accelerate.** Note that it was difficult to get DEMON to converge as β_{init} . needed to have small values;. bBigger β_{init} than 0.1 often diverged.

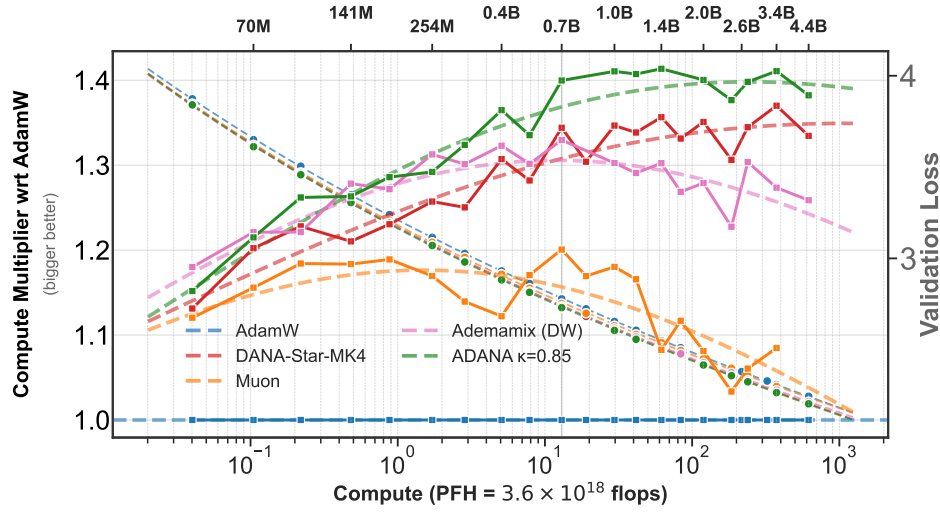


Figure K: **Scaling laws and compute multiplier vs compute C** for ADANA variants with transformers (Enoki model) on FineWeb. **Left axis:** compute-multiplier relative to ADAMW. ADANA and DANA-STAR-MK4 ($\kappa = 0.85$) benefits increase with scale; $\sim 40\%$ compute savings. ADEMAMIX (DW)⁰ (configured at $\kappa = 0.85$), ADANA ($\kappa = 0.75$), and MUON outperform ADAMW but with diminishing gains at larger scales; increasing κ to 0.85 improves scaling. **Right axis:** validation loss fit to a broken power-law $L = a + bC^{-c} + eC^{-f}$ with shared saturation a across optimizers.